Commodity-Currency Transmission Beyond a Single Pair: A Multi-Pair Replication

Hashim Khan

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Master of Mathematical Finance, University of Toronto

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Abstract

Commodity currencies move with their export commodity contemporaneously but, on the evidence of single-pair studies, do not forecast it. This paper asks whether that asymmetry generalizes, testing CAD against WTI, CLP against LME copper and NOK against Brent at daily frequency over 2010–2025, net of a broad-dollar and a broad-commodity control. It does. All three pairs show a strongly significant same-day association, in which a one-standard-deviation commodity move corresponds to between $+0.154\%$ and $+0.240\%$ of currency appreciation with Holm-adjusted p below 0.001, and none shows predictability at one or five days ahead, in sample, out of sample against a random walk, or under a walk-forward random forest. Every null is reported with a minimum detectable effect: the tests could have found partial R^2 of 0.29–0.43% and found 0.01–0.10%.

A fourth pair, AUD against iron ore, is reported separately as a methodological case study rather than as a comparison point. The SGX iron ore contract settles against the monthly average of a price-reporting-agency assessment, which attenuates the regressor by a factor that varies through the contract month and makes it the only positively autocorrelated commodity series in the study. Correcting for the attenuation does not rescue the pair: the corrected coefficient is an order of magnitude below the other three. Instrument choice, not geography, is the finding there.

The principal limitation is a placebo failure. Two non-commodity currencies load significantly on the commodity returns after the dollar control (sterling positively, the Swiss franc negatively), indicating a risk-appetite factor the dollar control does not absorb and which accounts for roughly 15–26% of the contemporaneous coefficients.

1 Introduction

Commodity-exporting economies are expected to see their currencies move with their export prices. If the link is causal and exploitable, commodity prices should forecast currency returns. If it is purely informational, with both assets repricing simultaneously to the same news, then it is contemporaneous only and forecastability is zero. The two worlds are indistinguishable in a correlation table and separable only with a lead–lag design.

The asymmetry is documented in the commodity-currency literature (Chen, Rogoff and Rossi 2010; Ferraro, Rogoff and Rossi 2015, the latter on oil and the Canadian dollar), and the specific finding replicated here is the soybean–BRL result from project 12 of this course.

The contribution is **breadth, not novelty**. The question is whether a pattern established mainly on single pairs holds across four economies with different export structures, different trading hours and different commodity market microstructures. A finding of "yes, in three of three testable pairs" is modest but useful; the fourth pair turns out to be more interesting for what it reveals about measurement than about Australia.

Half the contribution is a null. A null is only worth anything if the specification was fixed before the results were seen and the test had the power to find an effect. Section 3.4 and the accompanying `FREEZE.md` set out how that was enforced, and every null in this paper is reported alongside the smallest effect the test could have detected.

2 Data

2.1 Pairs and instruments

Table 1: Pairs, instruments and sources

Currency	Commodity	Instrument	Source
CAD	WTI crude	`CL1`, NYMEX front-month generic	RiskLab `RL-20930E28331E`
CLP	Copper Grade A	`LMCADS03`, LME 3-month forward	Bloomberg, student pull
NOK	Brent crude	`CO1`, ICE front-month generic	Bloomberg, student pull
AUD	Iron ore 62% Fe CFR China	`SCO1`, SGX front-month generic	Bloomberg, student pull

Controls are the ICE US Dollar Index (`RL-36290DAB949B`) and the Bloomberg Commodity Index (`RL-38259EF0EF46`). FX spot for AUD and CAD comes from the RiskLab G10 file (`RL-495F49CB82F5`); CLP and NOK were pulled directly.

Two instrument choices deserve a sentence. **Copper is LME, not COMEX**: Chile prices its exports off London, and COMEX fell 22% in a single session on 31 July 2025 on a US tariff dislocation absent from the LME, which would inject US trade policy into the Chilean regressor. **Commodities are futures, not spot**, because physical commodity "spot" is a once-daily agency assessment with no volume behind it, whereas futures have a documented exchange settlement time. FX is the reverse, since spot FX is the deep continuously traded market. The same principle therefore gives opposite answers on the two sides of the regression.

2.2 Provenance

Every raw input carries a SHA-256 hash and, where it came from the institutional library, a stable catalogue identifier. Four of the five catalogued files hash exactly to the catalogue's own recorded value; the fifth differs only because it is a format conversion of the catalogued CSV. Full records are in `manifests/source_inventory.csv` and `manifests/bloomberg_source_log.csv`.

The dollar control required correcting. A file named `USD_Index_DXY_Daily.csv` is *not* ICE DXY: it begins 2 January 2006 at 101.42 and reads 92.36 on 4 January 2010 where DXY was about 77.5, which identifies it as the Federal Reserve H.10 nominal broad dollar index, based at 100 in January 2006. Genuine ICE DXY was in the library under a different filename and is used instead; the Fed index is retained as a robustness control.

2.3 Sample and attrition

Table 2: Estimation samples after all exclusions

Pair	N (same day)	N ($h = 1$)	Window	Binding constraint
CAD	3,766	3,765	2010-01-05 – 2025-09-19	G10 FX file
CLP	3,629	3,628	2010-01-05 – 2025-09-23	BCOM
NOK	3,774	3,773	2010-01-05 – 2025-09-23	BCOM
AUD	1,023	1,022	2013-04-09 – 2025-09-10	contract history, §6
Pooled	11,169	11,151	n/a	CAD + CLP + NOK

The sample ends in September 2025 because two inherited files do, not because the Bloomberg pulls do; every directly pulled series reaches year end. Extending the window requires exactly two re-pulls, which were not obtainable within the study period (see Limitations).

Full stage-by-stage attrition is in `manifests/sample_attrition.csv` and the appendix table. The consequential exclusions are:

- **Roll days dropped** wherever the front-month contract pointer changes. A roll-day price change is a calendar spread, not a return anyone earned. Roll counts are CL 189, CO 193, SCO 152, all approximately twelve per year. `LMCADS03` is constant-maturity and shows zero, as it should.
- **WTI 20 and 21 April 2020 excluded.** A settlement of −$37.63 makes the log return undefined both into *and* out of the print. Affects CAD only.
- **Eighteen Chilean peso returns dropped** for spanning more than five calendar days, a consequence of a 2012 coverage hole of 121 missing weekdays.
- **Four Brent observations flagged** where the contract pointer is missing, so roll status is indeterminate rather than assumed.

3 Method

3.1 Specification

$$\Delta s_{t+h} \;=\; \alpha \;+\; \beta\,\Delta c_t \;+\; \gamma\,\Delta \mathrm{DXY}_t \;+\; \delta\,\Delta \mathrm{BCOM}_t^{\perp} \;+\; \varepsilon_{t+h} \tag{1}$$

with $h = 0$ contemporaneous, $h = 1$ the primary predictive horizon and $h = 5$ a prespecified secondary. All variables are daily log returns.

3.2 Sign convention

Every rate is expressed as **USD per unit of the commodity currency**, so an increase always means the commodity currency appreciated and β is expected positive in all four pairs. `AUDUSD` is already in that form; the other three are quoted locally per USD and their returns are negated, which is identical to inverting the price since $\log(1/x) = -\log(x)$.

The convention is verified two ways. Every convention-A return correlates negatively with the dollar factor, as it must (ρ from -0.32 to -0.70), and a worked example computed by hand for one pulled pair and one inverted pair is in `outputs/diagnostics/sign_convention_example.md`.

3.3 Controls

The broad-dollar control addresses the shared-numeraire problem: both sides of (1) carry a USD term, so a weaker dollar lifts both with nothing real happening. The quoting convention does not fix this; the control does.

The broad-commodity control is **orthogonalized**. BCOM holds the pair commodity directly (Brent at 10.58%, WTI at 8.85%, COMEX copper at 5.92%), so including raw BCOM would partial out part of the effect of interest. Instead BCOM is regressed on the pair's own commodity return and the residual is used, making β the *total* effect of an own-commodity move. Iron ore is not a BCOM constituent, so AUD needs no orthogonalization. Where the pair commodity is a constituent, the orthogonalization uses the instrument the index actually holds, which for copper is COMEX rather than LME.

3.4 Inference, and the specification freeze

Per pair: Newey–West HAC with the 1994 bandwidth rule and a normal reference. Pooled: pair fixed effects with Driscoll–Kraay errors.

The pooled estimator was amended before results were interpreted. The frozen specification called for two-way clustering by date and pair, which requires the cluster count to grow in both dimensions; the pooled panel has three pairs. A synthetic Monte Carlo (400 replications, $\beta = 0$ true) measured the frozen estimator's size at **0.5% against a nominal 5%**, which makes it invalid for this cluster count rather than merely conservative. Driscoll–Kraay came in at 4.5%, and its asymptotics ($T \to \infty$, N fixed) match a panel of three entities and 3,958 dates. The amendment changes no coefficient, only standard errors, and leaves both predictive nulls null. Both estimators are reported in Table 3.

Out-of-sample comparison uses **Clark–West**, not Diebold–Mariano. The model nests the random walk benchmark, since $\beta = \gamma = \delta = 0$ recovers it, and DM's asymptotics assume non-nested models. The brief specifies DM; this is a documented correction to it.

Multiple testing is handled by Holm across the six primary tests ({CAD, CLP, NOK} $\times$ {$h = 0$, $h = 1$}).

`FREEZE.md` records the SHA-256 of every specification file and every raw input at the specification freeze. Seven decisions, their reasoning and every subsequent amendment with its justification are in `design_decisions.md`.

4 Results

4.1 Same day

All three pairs show a strongly significant contemporaneous association, surviving both controls and Holm adjustment.

Copper transmits roughly three times as strongly per unit as crude does to either oil currency, which is what the export structures suggest: copper is approximately half of Chilean goods exports, whereas crude is a large but smaller share for Canada and Norway.

Table 3: Contemporaneous transmission, $h = 0$

Pair	β	1-SD commodity move → FX	Partial R^2	Holm p
CAD	+0.0613	+**0.154**% (WTI 1 SD = 2.51%)	14.5%	< 0.001
CLP	+0.1800	+**0.240**% (copper 1 SD = 1.33%)	10.0%	< 0.001
NOK	+0.0984	+**0.217**% (Brent 1 SD = 2.21%)	16.2%	< 0.001
Pooled	+0.0889	+0.185%	9.9%	< 0.001

Full table, including standard errors, confidence intervals, both pooled estimators and all $h = 5$ rows: `outputs/tables/table1_in_sample.md`.

4.2 Next day

Nothing, by four independent routes.

In sample. No pair is significant at $h = 1$ (p from 0.106 to 0.661; lowest Holm-adjusted p is 0.318) or at $h = 5$.

Power. The tests were not weak. At 80% power they could have detected partial R^2 of 0.43% (CAD), 0.29% (CLP) and 0.37% (NOK); what they found was 0.01%, 0.10% and 0.02%, an order of magnitude below the detection threshold in every case.

Out of sample. No pair beats a driftless random walk at either horizon. Out-of-sample R^2 is negative or zero for five of six specifications, and the lowest Clark–West p-value across all of them is 0.091 (CLP, $h = 1$). The MAE ratio never falls below 1.000.

Under a nonlinear model. A walk-forward random forest with access to squares and lags of every regressor, hyperparameters fixed and untuned, beat the random walk in 2 of 14 specifications (both AUD at $h = 5$, both by under 0.4% OOS R^2), and **no specification produced a deflated Sharpe ratio above 0.95** (highest 0.318). Diebold–Mariano is legitimate here, and only here, because the forest does not nest the benchmark; no specification approaches significance (lowest $p = 0.934$).

That last result matters more than simple agreement would. It rules out the first objection to a linear predictive null: that the null is an artifact of assuming linearity.

4.3 After costs

The baseline specification is a regression. It estimates an association and takes no position, so transaction costs do not enter it. The random forest is different: it converts each forecast into a signed position and reports a Sharpe ratio, and a Sharpe ratio before costs is not a Sharpe ratio. Costs are therefore applied to the forest and to nothing else.

They are *declared* rather than estimated. No usable bid–ask series exists in the data (the one bid column available is stale and unusable), so inventing one from it would be worse than stating an assumption plainly. The assumption is a one-way half-spread of 1.0 basis point for CAD and AUD, 1.5 for NOK and 7.5 for CLP, tripled through the 2020 H1 stress window used elsewhere for the influence check. These are conservative round numbers for daily interbank FX, and the emerging-market cross is deliberately charged several times the majors. Cost is levied on the *change* in position, so holding is free and a sign flip pays a full round trip.

Costs cut the number of specifications with a positive Sharpe from 10 of 14 to 4, and **none** of the fourteen clears a deflated Sharpe of 0.95 once charged; the highest is 0.184. CLP is the extreme case, losing roughly 18% a year to turnover alone, which is what a 7.5 bp spread

Table 4: The random forest strategy before and after costs

Specification	Turnover per day	Cost %/year	Sharpe gross	Sharpe net	DSR net	Break-even half-spread, bp
CAD, $h = 1$	1.00	2.8	+0.01	−0.41	0.002	0.02
CAD, $h = 5$	0.70	2.0	+0.16	−0.14	0.017	0.61
CLP, $h = 1$	0.81	17.6	+0.39	−0.86	0.000	2.68
CLP, $h = 5$	0.87	18.2	+0.21	−1.09	0.000	1.35
NOK, $h = 1$	0.87	3.8	−0.18	−0.50	0.001	n/a
NOK, $h = 5$	1.04	4.4	+0.06	−0.31	0.005	0.28

Annualised Sharpe. DSR is the deflated Sharpe ratio against 14 trials. Break-even is the one-way half-spread at which the mean net return reaches zero, and is undefined where the gross mean is already negative. The eight AUD case-study specifications behave the same way: gross Sharpe −0.97 to +0.70, net −1.19 to +0.45, highest net DSR 0.184, break-even 0.27 to 3.54 bp. Full table: `outputs/tables/results_costs.csv`.

does to a strategy that changes position most days.

The number that does not depend on my assumption is the *break-even half-spread* in the last column: the cost at which the strategy stops making money, computed directly from gross return and turnover. Where the gross Sharpe is positive at all, it lies between 0.02 and 3.54 bp. CAD at $h = 1$ breaks even at two hundredths of a basis point, which is to say it never had anything to give away. Even the most favourable case, an AUD specification at 3.54 bp, demands execution a desk would not achieve on the pairs where the gross signal is largest. Readers who prefer tighter or wider spread assumptions than mine can apply them to that column directly.

Carry and financing are **not** computed. Daily carry is the interest differential, which requires forward points; those were never obtained (see Limitations). The omission is declared rather than approximated. Its magnitude over a one-day holding period is small relative to the spreads above, but its sign is unknown, because it depends on whether the strategy happens to sit long or short the higher-yielding currency on any given day, and over this sample that varies.

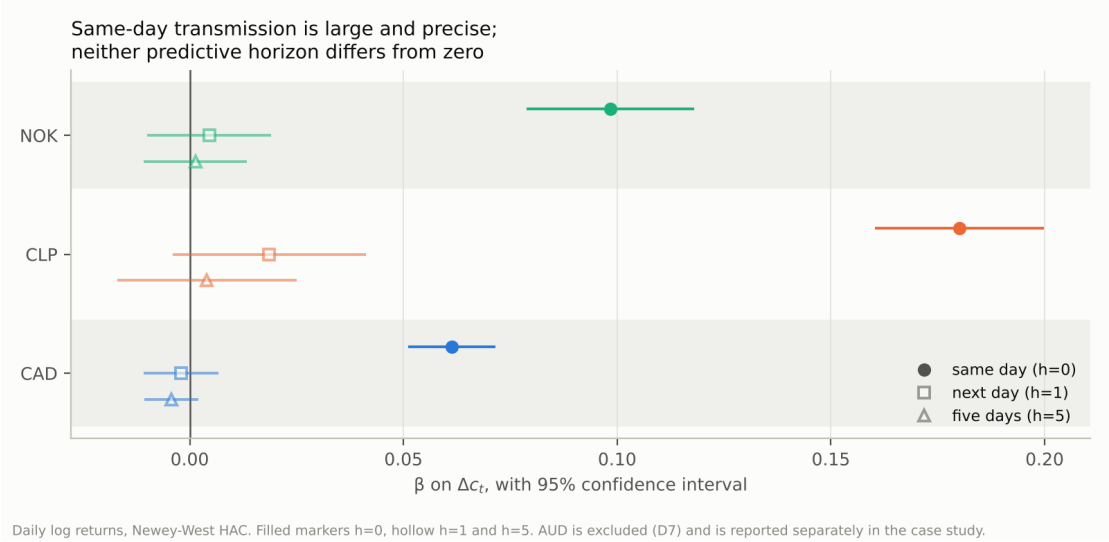


Figure 1: Coefficients with 95% confidence intervals at all three horizons. The same-day estimates sit far from zero with tight intervals; both predictive horizons straddle it.

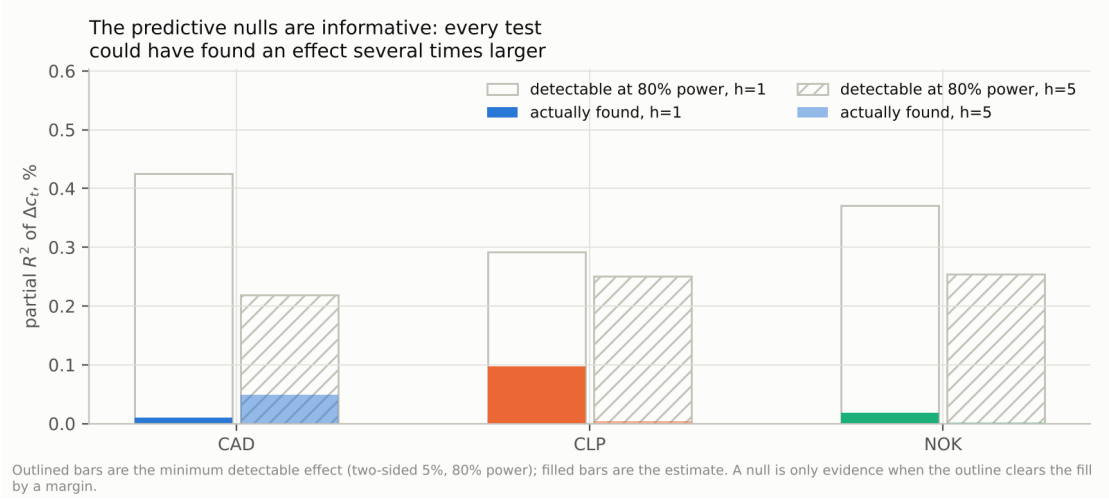


Figure 2: Minimum detectable effect against the estimate actually obtained. A null is evidence only when the outlined bar clears the filled one by a margin, which it does at both predictive horizons for all three pairs.

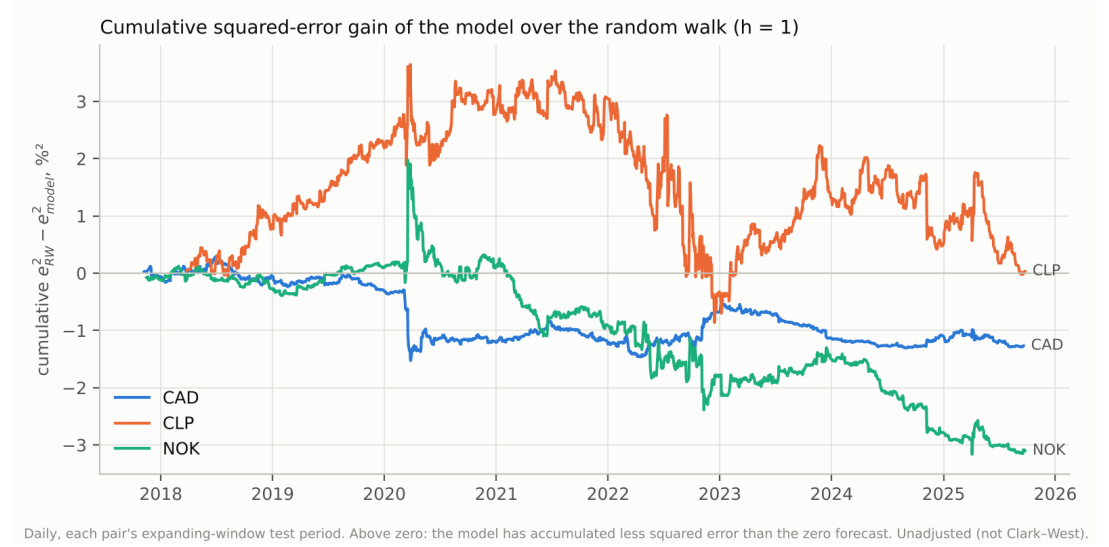


Figure 3: Cumulative squared-error loss differential against the random walk. A persistently rising line would indicate genuine predictive gain; none is present at either horizon for any pair.

4.4 The placebo fails, and this is the principal limitation

Two non-commodity currencies were used as negative controls. Neither Switzerland nor the United Kingdom exports these commodities, so neither should respond to them once the dollar is controlled. Both do.

The pattern is a **risk-appetite factor the dollar control does not absorb**: risk-on lifts commodities and commodity currencies strongly, lifts a cyclical non-commodity currency weakly, and weakens the canonical safe haven. Sterling loads positive because it shares the factor; the franc loads negative because it mirrors it. The franc alone would have been a poor control for exactly this reason, which is why sterling was added.

Taking the sterling loading as the generic cyclical-currency response, the common factor accounts for roughly **26% of the CAD coefficient, 15% of CLP's and 20% of NOK's**. The remainder is plausibly commodity-specific, and the sign, significance and ordering of the three

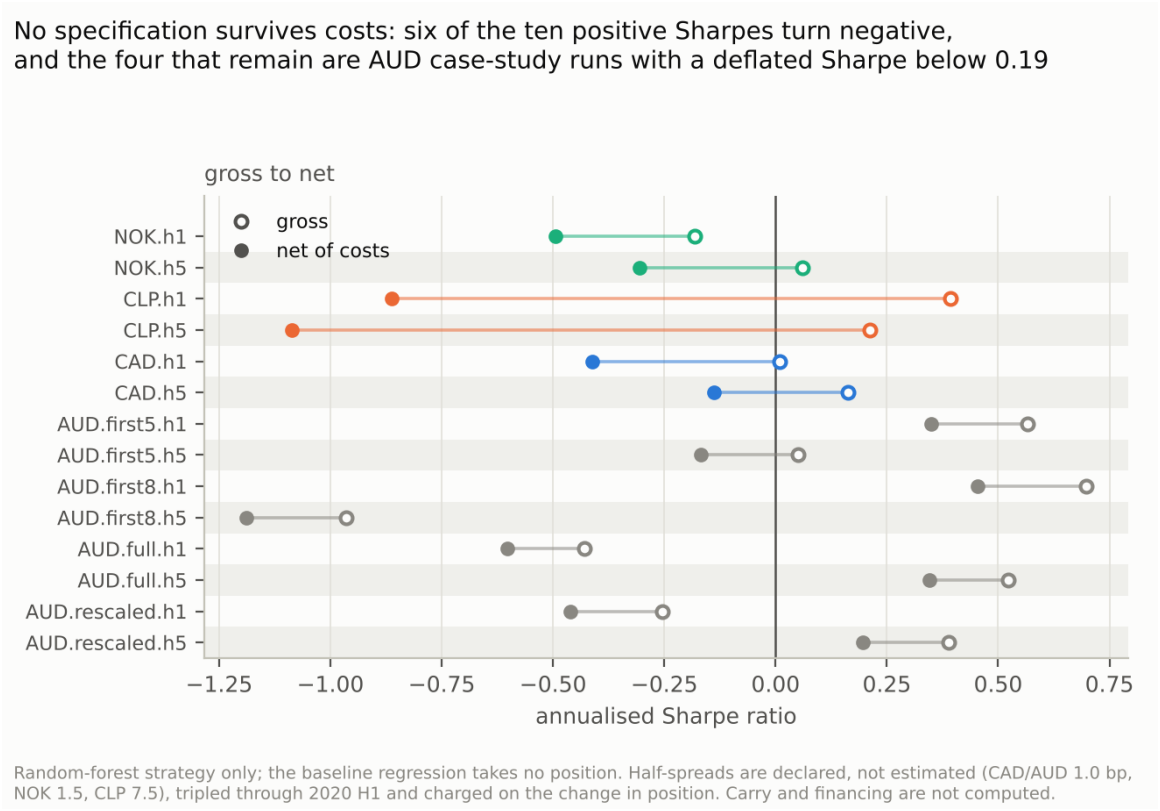


Figure 4: Costs on the random-forest strategy, by specification: the move from gross Sharpe (hollow) to net (filled). A line crossing the zero rule marks a strategy that only made money before paying for itself. The four that stay positive are all AUD case-study specifications, and none of them clears a deflated Sharpe of 0.95.

Table 5: Placebo outcomes, $h = 0$

	CAD	CLP	NOK
Commodity currency β	$+0.0613$	$+0.1800$	$+0.0984$
GBP placebo	$\mathbf{+0.0158}$	$\mathbf{+0.0269}$	$\mathbf{+0.0193}$
CHF placebo	$\mathbf{-0.0198}$	-0.0157	$\mathbf{-0.0209}$
GBP p	< 0.001	< 0.001	< 0.001
CHF p	< 0.001	0.052	< 0.001

pairs are unaffected. But the contemporaneous result cannot be read as a pure terms-of-trade effect, and it is not reported as one here.

The predictive null is untouched: a common contemporaneous factor does not create forecastability, and none is found.

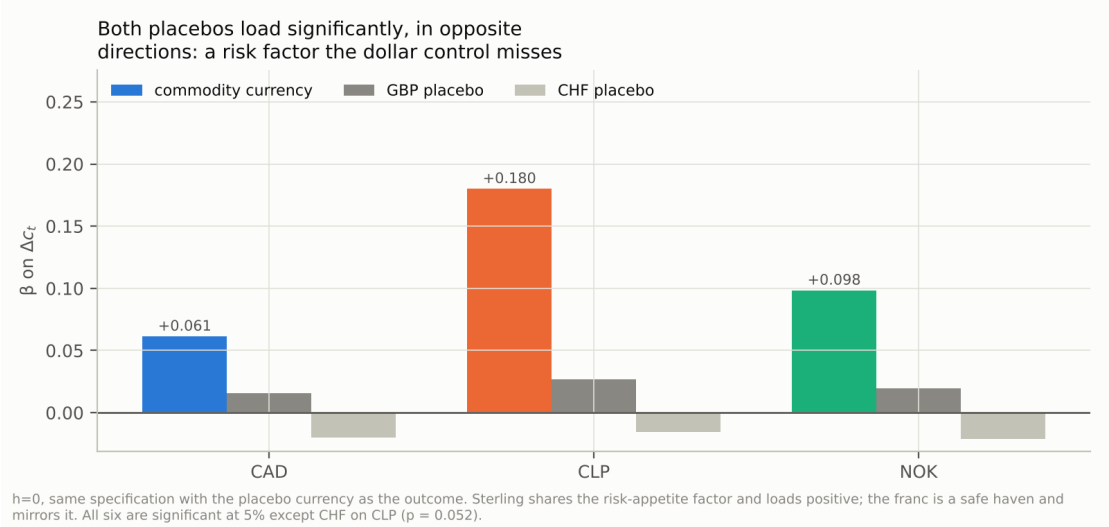


Figure 5: Placebo loadings beside the commodity-currency coefficients. Sterling shares the risk-appetite factor and loads positive; the franc is a safe haven and mirrors it. Both are an order of magnitude smaller than the effect of interest.

5 Robustness

Table 4 of the output set reports every prespecified check. Nothing overturns the headline results; three are worth noting.

The dollar control matters most for CLP. Replacing ICE DXY with the Fed broad dollar index moves the CLP coefficient from 0.180 to 0.114, the largest sensitivity anywhere in the study. The Fed index is weighted toward trading partners including China, so it plausibly absorbs part of the Chinese demand factor that drives copper. The other pairs barely move.

CAD is a DXY constituent. The dollar control contains the dependent variable at 9.1% weight for that pair alone. A dollar index rebuilt without CAD from its remaining components (covering 86.7% of the non-CAD basket; SEK unavailable) correlates 0.953 with the full index and leaves the coefficient essentially unchanged.

The WTI source question closed empirically. Two library files disagreed on one date. Retrieving the alternative showed the two are **identical on 3,960 of 3,961 overlapping dates**, differing only on 23 September 2025 by 0.28. Re-estimating CAD on the alternative reproduces the coefficients to four decimal places.

Also run without material effect: BCOM orthogonalized on LME rather than COMEX copper; NOK restricted to pre-2022, when Brent still represented Norwegian export revenue before gas became comparable; long-span Chilean returns retained with a span control; indeterminate-roll Brent observations retained; 2020 H1 dropped; the largest commodity move dropped per pair.

One check could not be run: the AUD zero-return diagnostic, because the frozen specification defined its trigger ("if AUD proves unstable at the sample start") without an operational rule.

That is recorded as an omission rather than quietly dropped.

6 AUD and iron ore: a case study in instrument choice

AUD is reported separately and is not part of the cross-pair comparison. The reason is not that it gives an inconvenient answer but that its regressor is measured differently from the other three in a way that is derivable in advance.

6.1 The mechanism

The SGX contract settles against the **arithmetic average** of all IODEX 62% Fe prints in the expiring month. With N prints in the month and k elapsed, a shock to today's index enters the realized portion at weight $1/N$ and the forecast of the remaining prints at weight $(N-k)/N$, so under a random walk the pass-through from index to futures is

$$w_k = \frac{N-k+1}{N} \tag{2}$$

falling from approximately 1 on the first day to $1/N$ by the last. The data agrees: mean absolute daily return falls from 1.36% on day 2 to 0.17% on day 21, while WTI and Brent remain flat at 1–2% throughout their contract months.

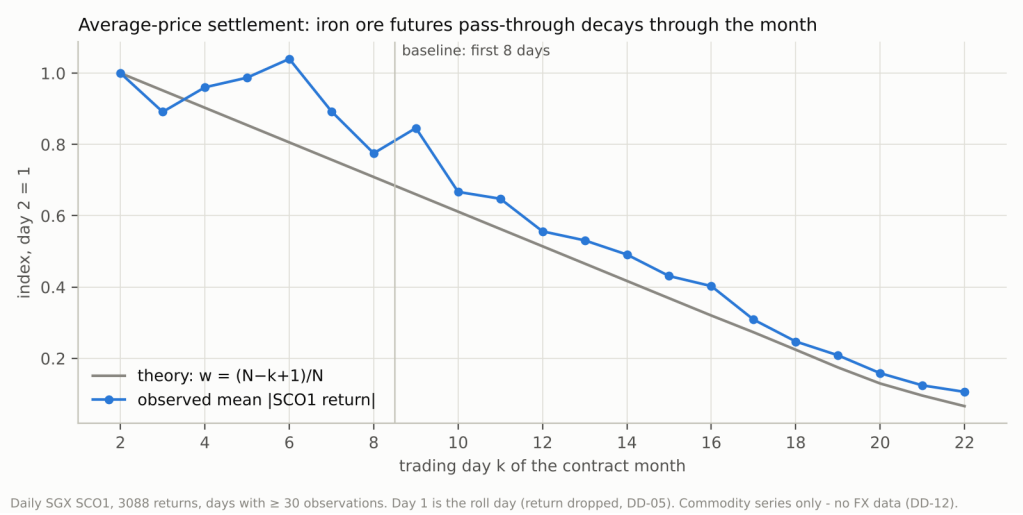


Figure 6: Mean absolute daily return by trading day within the contract month. Iron ore decays monotonically, consistent with (2); crude does not.

A second property compounds it. Agency assessments are sticky, since values are carried forward when physical liquidity is thin, and iron ore is the only commodity in the study with **positive** first-order autocorrelation.

Crude and copper show the slight negative autocorrelation characteristic of bid–ask bounce. A positively autocorrelated regressor manufactures a lead–lag artifact, and AUD is the one pair that fails the date-shift alignment test: its coefficient at a one-day backward shift (+0.0227, $t = 2.51$) *exceeds* its contemporaneous coefficient (+0.0166). The two symptoms share one cause.

Table 6: First-order autocorrelation of commodity log returns

Series	AC(1)	t	Zero-return days
Iron ore	$+\mathbf{0.046}$	$+2.57$	**2.12**%
WTI	-0.033	-2.03	0.42%
Brent	-0.006	-0.39	0.25%
Copper	-0.031	-1.99	0.89%

6.2 The correction does not rescue the pair

Table 7: AUD under four treatments of the iron ore regressor

Treatment	β	p	N
Full sample	$+0.0166$	0.016	2,963
First 8 days of contract month	$+0.0166$	0.037	1,023
First 5 days	$+0.0125$	0.235	575
Rescaled by $1/w$	$+\mathbf{0.0075}$	0.047	2,963

The rescaling behaves exactly as the algebra predicts, which confirms the correction is working. But it puts the attenuation-corrected coefficient at roughly 0.0075, against 0.061 for CAD and 0.180 for CLP, an order of magnitude below rather than comparable.

So the measurement problem is real, documented, and not what makes AUD weak. Two explanations remain and neither is tested here. The first is timing: on published and conventional settlement times, iron ore settles approximately 10.5 hours before the FX snap, against 2.5 hours for the oil pairs, and the gap moves on 37.8% of days because Singapore observes no daylight saving while New York does. The same-day AUD observation therefore contains most of a trading day *after* the commodity print. The second is simply that iron ore transmits less strongly to AUD than oil does to CAD and NOK or copper does to CLP, which is plausible given that coal is comparable to iron ore in the Australian export basket.

The useful result here is a negative one about method: a correction derived from first principles, applied correctly, and found not to explain the anomaly it was built for.

7 Limitations

Stated in descending order of how much they constrain the conclusions.

1. **The placebo failure** (§4.4). A risk-appetite factor the dollar control does not absorb accounts for roughly 15–26% of the contemporaneous coefficients. The same-day result is not a pure terms-of-trade estimate.
2. **AUD carries two independent measurement problems** and is excluded from the comparison for that reason, leaving a three-pair rather than four-pair test.
3. **The sample ends September 2025**, three months short of the requested window, because of inherited file coverage.

4. **The Chilean peso has a 2012 coverage hole** of 121 missing weekdays; 18 returns were excluded for spanning more than five calendar days.
5. **The iron ore ticker was recovered from returned contract codes**, not confirmed on the terminal.
6. **No volume or open-interest data** was obtainable, so early-contract liquidity could not be verified.
7. **Bloomberg Desktop API access was suspended**, which prevented the sample extension.
8. $h = 5$ **is the single-day return five days ahead**, as the frozen equation defines it, not a cumulative five-day return. The cumulative version was not tested.

8 Conclusion

The contemporaneous-strong, predictively-weak pattern generalizes. Across CAD, CLP and NOK, a one-standard-deviation commodity move corresponds to between $+0.154\%$ and $+0.240\%$ of same-day currency appreciation, significant beyond any conventional threshold after controlling for broad dollar and broad commodity moves and after Holm adjustment. None of the three forecasts the next day, and the nulls are informative rather than merely quiet: the tests could have detected effects several times larger than those observed, no pair beats a random walk out of sample, and a nonlinear learner finds nothing the linear model missed.

Two qualifications carry real weight. The first is that a risk-appetite factor survives the dollar control and accounts for perhaps a fifth of the contemporaneous relationship, so the same-day finding is not purely a terms-of-trade effect. The second is the fourth pair: AUD could not be included, because the only available iron ore instrument settles against a monthly average of a smoothed index, and correcting for that, which can be done and was, leaves the coefficient an order of magnitude below the others rather than in line with them.

That second qualification may be the more transferable finding. A literature built on commodity-currency transmission takes the commodity price series as given. For iron ore there is no unattenuated traded instrument: every cleared contract settles on a monthly average, because that is how the physical market contracts. Any study including AUD inherits that, whether or not it says so.

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Reproducibility

Specification freeze	26 October 2026, hashes in `FREEZE.md`
Decisions and amendments	`design_decisions.md` (D1–D7)
Source provenance	`manifests/source_inventory.csv`, `bloomberg_source_log.csv`
Variable definitions	`manifests/variable_dictionary.csv`
Sample attrition	`manifests/sample_attrition.csv`
Forecast origins	`manifests/forecast_origins.csv`
Machine-readable results	`outputs/tables/results_primary.csv`, `results_robustness.csv`, `results_rf_dsr.`
Seed	20261026
Rebuild	`src/00` → `01` → `03` → `04` → `05` → `06` → `07` → `08` → `08b` → `08c` → `09`

Every table and figure in this paper is generated by `src/09_tables_figures.py` from the saved results tables. None was edited by hand.